

Evaluating the "Resilience Premium" in New Jersey Municipal Bond Markets 2015-2025

A. Research Objective

The primary objective of this project is to quantify the relationship between climate risk, adaptation measures, and municipal borrowing costs in New Jersey. Specifically, the research aims to determine:

1. **The SLR Penalty:** Does the market demand higher yields (spreads) for bonds issued by "high-exposure" coastal municipalities compared to inland towns with identical credit ratings?
2. **The Resilience Premium:** Do coastal municipalities that have undertaken significant adaptation, defined by NJ I-Bank funding or high FEMA Community Rating System (CRS) scores, recover some of this penalty through lower borrowing costs?

This research will provide the NJDEP with a data-backed narrative to demonstrate the Return on Investment (ROI) of climate resilience projects to local treasury officials and taxpayers.

B. Method

The project will utilize a Matched-Pair Econometric Analysis and a Difference-in-Differences (DiD) model to isolate the financial impact of adaptation.

1. **Categorization of Municipalities:**
 - **Exposure Baseline:** Using NJ FloodMapper data, municipalities will be categorized by the percentage of their tax base vulnerable to 2.1ft, 4.1ft, and 6.1ft Sea Level Rise (SLR) to align with NJ PACT 2100 projections.
 - **Adaptation Variable:** Municipalities will be scored based on their FEMA CRS Class (1–10) and their participation in the NJ Water Bank (I-Bank), specifically looking for the 150 priority points awarded for "Climate Resilience Measures" since SFY 2023.
2. **Financial Modeling:**
 - **Yield Spread Calculation:** For each municipal bond issued between 2015 and 2025, we will calculate the "Spread over AAA," defined as the difference between the bond's offering yield and the MMD (Municipal Market Data) AAA Benchmark for the same maturity on the date of issuance.
 - **Regression Analysis:** We will regress these spreads against climate exposure while controlling for standard credit factors (Debt-to-GDP, Median Household Income, and Credit Rating).
3. **The "2022 Pivot" Analysis:** A specific sub-study will analyze if the introduction of NJDEP's climate resilience ranking points in 2022 led to a measurable compression in spreads for projects labeled as "Resilient" versus those that were not.

4. **Controls for Bond Structure:** We will control for Bond Insurance (Insured vs. Uninsured), Issue Size (Liquidity proxy), and Tax Status (AMT vs. Non-AMT) to ensure yield spreads reflect credit risk rather than structural market factors.

C. Data Sources

- **Climate Risk:** NJ FloodMapper (inundation parcels); NJDEP Scientific Report on Climate Change ((NJ PACT SLR projections).
- **Adaptation Activity:** NJ I-Bank Intended Use Plans (IUPs) (Project Priority Lists and ranking points); FEMA National Flood Insurance Program (NFIP) (Community Rating System scores).
- **Financial Data:** MSRB EMMA (CUSIP-level yield at issuance, maturity, and risk disclosure text); Bloomberg/TM3 (AAA Municipal Market Data benchmarks).
- **Economic Controls:** NJ Department of Community Affairs (DCA) (Municipal budget and tax base data).

D. Project Timeline

- **Weeks 1–2: Data Ingestion:** Scrape EMMA filings for coastal NJ bond issuances and join with NJ FloodMapper exposure data.
- **Weeks 3–4: Classification:** Define "Resilient" vs. "Non-Resilient" cohorts based on I-Bank funding history and CRS rankings.
- **Weeks 5–7: Econometric Modeling:** Run the DiD and Matched-Pair regressions to calculate the "SLR Penalty" and "Resilience Premium."
- **Weeks 8–9: Final Synthesis:** Draft the technical report and create visualizations showing the interest savings (in dollars) for resilient towns.

E. Deliverables

- **Technical Paper:** A formal report detailing the basis point (bps) delta between prepared and unprepared communities.
- **Presentation:** A slide deck summarizing the project's methodology, key findings, and strategic implications for state and local planners.
- **Data Tables:** A spreadsheet for the NJDEP based on our data and analysis.